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RESEARCH ARTICLE



The impact of pensions on the income gap among the elderly: the case of Taiwan's National Pension

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ABSTRACT

In most countries, first-pillar public pension systems provide comprehensive coverage for all citizens. By contrast, Taiwan's unique National Pension (NP) is a compulsory, defined-benefit pension insurance that covers citizens excluded from occupation-based social insurance. As such, the NP inevitably includes disadvantaged citizens in its coverage. Using NP administrative data to link labour insurance, social welfare allowance, and income tax, we find that the NP narrows the pension benefit gap between the disadvantaged and the non-disadvantaged. The narrowing effect is stronger for the younger than for the older generation. Furthermore, the NP enhances pension benefits for women more than for men, thereby narrowing the difference in pension benefits between the sexes. As expected, the NP is more important for low-income citizens than for their wealthy counterparts.

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1. Introduction

Population ageing has become a nearly universal concern in countries around the world. Based on United Nations data, older persons accounted for 20% of the global population in 2023, and are expected to reach 27.8% by 2050 (United Nations Department of Economic and Social Affairs, Population Division, 2023). As a consequence, public policies to care and provide for the growing elderly population have posed severe challenges to countries around the world (Hinrichs, 2021). Old age protection has become the most common social security programme around the world, with 166 countries implementing such programmes by the end of the 20th century (US Department of Social Security Administration, 1997). This study focuses on older people's basic income support system to evaluate its impact on alleviating poverty among the elderly, highlighting the potential for public policies to make a significant difference in the lives of older people.

Public pension systems can be categorized into two different models, namely, the Beveridge and the Bismarck models (Ebbinghaus, 2021; Kolmar, 2007; Palier & Bonoli, 1995). The former provides universal protection to the whole population in a country,

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whereas the latter usually covers only the employed population. It is argued that contributory Bismarckian systems in some countries may be better suited to reducing old-age poverty (Ebbinghaus, 2021). However, the Bismarck pension system faces an obvious challenge in that it often fails to provide adequate old-age income protection to those who are not employed.

Countries with Bismarckian pension systems employ different strategies to enhance old-age protection for those who are not employed. Many of them provide tax-funded old-age benefits to those who do not receive sufficient pension benefits, subject to means-testing. Taiwan's National Pension (NP) is a unique case in the world in that it provides an old-age pension to those who are not employed based on a contributory principle. Due to its uniqueness, and in light of the pressing global demographic challenge, the impact of Taiwan's NP on reducing old-age poverty is worthy of thorough research.

Taiwan's NP is a typical Bismarckian model comprising four main schemes. Three contributory schemes cover public-sector employees, private-sector employees, and working-age individuals who are not employed: (1) Government Employee and School Staff Insurance, (2) Labor Insurance (LI), and (3) the NP, respectively. The fourth, Older Farmers' Allowance, is a tax-funded benefit paid to older people who have worked in the agricultural sector. Under these arrangements, the state pension system covers the entire working-age population. Of these four schemes, the LI is the most extensive, covering 10.4 million participants, comprising 68.1% of the population aged 20 to 65 in 2022 (Bureau of Labor Insurance, 2022a). The NP is the second largest scheme, with 2.8 million participants in 2022, comprising 14.4% of the population aged 20 to 65 in 2022 (Bureau of Labor Insurance, 2022b).

In addition to the four schemes introduced above, there are social welfare allowance (SWA) programmes that benefit older people in economically disadvantaged groups. For instance, the state provides means-tested benefits to older people living on a low income. Supplementary benefits are provided to those whose social insurance pension benefits are below a particular threshold. In addition, veterans are eligible for a tax-funded benefit (also subject to means testing). These programmes are limited in terms of their coverage and benefits.

Taiwan's NP differs from those of Japan and South Korea. In Japan, all persons aged 20 to 59 participate in the NP, which provides a 'Basic Pension' on the basis of old age, disability, or death. Korea's National Pension Service was established to provide citizens working in various occupational groups with income security through national welfare benefits for retirement, disability, or death. By comparison, Taiwan's NP, established in 2008, only covers citizens aged 25 – 64 who do not participate in any occupational-based pension schemes. Citizens who fall into this group must join this programme.

Like the schemes in Japan and Korea, Taiwan's NP faces the ageing-related fiscal challenges of defined-benefit pension systems. Kudrna et al. (2022) consider how various countries have mitigated the pressing fiscal cost of old-age public pension programmes and redistributed pension benefits to those in need with shorter life expectancies. To this end, countries such as Australia, Denmark, and the UK have pension systems in which the benefits to older eligible citizens are dependent on their economic status, with means testing based on current income and assets independent of past labour earnings. Kudrna et al. (2022) investigate the extent to which this built-

in mechanism can contain the adverse effects of population ageing on a pension system's fiscal costs and progressivity. Taiwan's NP is similar to these new programmes in that it assists citizens with low income; rather than directly enhancing their pension benefits, it subsidizes their NP premiums. With NP's defined-benefit feature, low-income citizens may be entitled to their pension benefit at a lower premium.

The main purpose of Taiwan's NP is to improve income protection for older people in socioeconomically disadvantaged groups. More than a decade after implementation, it remains unclear whether the NP has achieved its policy purpose. More specifically, what are the impacts of the NP on the economic condition of citizens from different socioeconomic statuses? Do its benefits narrow the difference in economic conditions between disadvantaged and non-disadvantaged groups? These are the issues we explore in this study.

The remainder of the study is organized as follows. In [Section 2](#), we review the related literature, and in [Section 3](#), we introduce the design of Taiwan's NP. In [Section 4](#), we set out the demographic profile of NP participants and their premium-paying behaviours. In [Section 5](#), we compare the pension benefits across socioeconomic groups and generations, calculating the actual pension benefits of the NP and the SWA for citizens older than 65; we also predict the NP and LI pension benefits, as well as SWA for citizens younger than 65. In [Section 6](#), we use the predictions from [Section 5](#) to conduct a regression analysis of the relationships between different future pension benefits and demographic characteristics, including sex, age, income, and degree of disadvantage. In [Section 7](#), we conclude and present the policy implications of our study.

2. Literature review

2.1. Pension schemes and income equality

Few studies have investigated pension schemes' effect on equity, and although some have emerged in the last decade, most have been theoretical. For example, Jensen et al. (2021) divided citizens into long- and short-longevity groups, concluding that designing a pension system based on average life expectancy hurts those in the short-longevity group. They also found that the benefit to those with greater longevity increases as more premiums are paid by those in the short-longevity group. Similarly, Auerbach et al. (2017) found that citizens in higher-income groups obtain more pension benefits than those in low-income groups because the former tend to have longer life expectancies. Andersen (2014) previously reached a similar conclusion.

Laun et al. (2019) developed a theoretical model with heterogeneities of mortality to investigate pension reforms, such as increasing the early access age to retirement, raising income taxes, and lowering old-age retirement and disability benefits. The authors used educational attainment to proxy socioeconomic status and found that improvements in longevity benefit the highly educated more than their less educated counterparts. Ayuso et al. (2017) modelled a hypothetical social security system, arguing that heterogeneity in longevity diminishes the effects of pension reform. They suggest developing mechanisms to reduce the impact of this heterogeneity through pension design.

Sanchez-Romer et al (Sanchez-Romero et al., 2020). used an overlapping generation model to analyse the welfare of those with high and low socioeconomic status in the

pension system, finding that differences in the groups' life expectancies cause those in the low-income group to suffer welfare losses while the high-income group enjoys welfare gains. In sum, heterogeneous longevity is the crucial factor generating inequality in pension benefits. It is well known that the disadvantaged have shorter lives than the non-disadvantaged.

Very few studies have used real data to address the redistribution effect of public pension systems. Hsieh and Tung (2016) may have been the first to probe the impact of Taiwan's NP on inequality. They used aggregate data and applied the general accounting method, concluding that the introduction of the NP reduced intergenerational inequality. Chang et al. (2023) used Taiwan's NP administrative data to investigate the sustainability of the NP and inequalities across groups. They showed that, although inter- and intragenerational inequalities are reflected in the NP, these are mitigated by the premium subsidy provided by Taiwan's government. The NP subsidy cited in this study refers to that premium subsidy.

2.2. Pension models and old-age poverty

The Bismarck model, initially adopted in Germany, is also referred to as the 'Continental model' (Scheubel, 2013). The Bismarck model, which has been implemented in most European countries,¹ values income smoothing above other pension system characteristics (Hinrichs, 2021; Krieger & Traub, 2013; Matković & Stanić, 2020). 'Income smoothing' refers to the process of redistributing income over a person's lifetime to ensure a consistent standard of living. By such means, the Bismarck model is designed to ensure a retirement income that is strongly related to pre-retirement earnings. Because the level of pension benefits is determined by individuals' contributions based on their actual wages, the Bismarck model is also called the 'social insurance model' (Scheubel, 2013). Due to its emphasis on maintaining individuals' standards of living, the Bismarckian system has a limited effect on income redistribution – and thus, theoretically, income inequality is preserved among the elderly (Klitzke, 2017).

The Beveridge model, also referred to as the 'Anglo-Saxon model', began in the United Kingdom (Scheubel, 2013). It has been adopted in some European countries such as Denmark, Finland, Ireland, the Netherlands, Sweden, and the United Kingdom (Assal et al., 2023). The Beveridge model employs a flat-rate basic pension financed primarily by taxes or tax-like contributions (Hinrichs, 2021). The Beveridgean system aims to provide a minimum pension to avoid elderly poverty. A Beveridgean system usually implies a two-tiered pension system, with the first tier being a flat-rate benefit to prevent poverty and the second tier operating through occupational pensions or private pensions to maintain economic status and consumption smoothing (Hinrichs, 2021; Klitzke, 2017).

In addition, countries with a Beveridgean system have developed more mature private pension systems, and private pension plans are more prevalent there than in countries with a Bismarckian system. This implies that a Bismarckian system may impede the development of private pensions (Ebbinghaus & Möhring, 2022). As a result, compared with a Bismarckian system, a Beveridgean system requires lower contributions and better accommodates citizens who demand greater pension benefits from their income using private insurance. Therefore, countries adopting

a Beveridgean system have lower pension expenditures in terms of GDP than countries adopting a Bismarckian system (Conde Ruiz & González, 2016; Gough & Adami, 2012).

Alongside the trend of ageing populations, women's participation in the labour force also continues to rise. This phenomenon puts more demographic pressure on countries with Bismarckian systems than those within Beveridgean systems (Schludi, 2005) because the latter grants universal flat-rate benefits. Finally, although the Beveridge model is designed to prevent elderly poverty, as Ebbinghaus (2021) argued, in practice, countries with a generous replacement rate in a Bismarckian system may be more successful in preventing elderly poverty than countries with a Beveridgean system.

Taiwan has adopted the Bismarckian system, providing pensions to retirees based on their contribution years in the labour market. Before 2008, citizens who were not employed were excluded from this system and did not receive old-age protection. However, the implementation of the NP completed the coverage of the entire Taiwan population. This study seeks to investigate the effect of the NP on the reduction of the income gap among the elderly, especially the disadvantaged. We use Taiwan's NP administrative data and link it with data from the LI and old-age SWA, as well as information on individual income from administrative tax data. This merged data set offers a complete picture of the social safety net in Taiwan, capturing citizen movement between the NP and the LI.

This article contributes to the literature on public pension programmes in at least five ways. First, because of a lack of available data, most extant studies use theoretical models to explore the impact of pension systems on the income gap between socioeconomic groups; we examine this issue using empirical administrative data. Second, our data contain information on all specified disadvantaged groups (Table 1), which allows us to explore how the NP impacts the income gap among these groups. Third, our data reflect complete information on the social safety net available to citizens, including NP, LI pension, and old-age SWA. The richness of the data enables us to explore how the NP impacts the income gap among disadvantaged groups participating in the NP and the social safety net more broadly. Fourth, this study provides valuable insights into Taiwan's unique social insurance system, which is separate from other occupational-based pension schemes and is designed to protect the elderly disadvantaged population. Fifth, this research seeks to explain how the unique design of Taiwan's NP works to reduce the income gap among the elderly; compared to other countries, Taiwan's NP does not provide more direct benefits to the poor, but rather indirectly subsidizes more premiums to them.

3. Design of Taiwan's National Pension

Taiwan's NP is a compulsory scheme in that Taiwanese citizens not participating in any occupation-based pension schemes are required to join. It is a defined-benefit scheme financed by participant contributions, with subsidies provided by the government. The current contribution rate is 10% of insured income. Notably, NP participants are not supposed to earn wages from employment. The monthly insured amount (MIA) for contributions to the NP was set as the minimum wage when the NP was first implemented in October 2008; this has since been indexed with consumer prices. Since January 2023, the MIA for calculating contributions and benefits has been set at NT\$18,282.

It is also important to note that NP participants are working-age citizens with no employer who shares their defined contribution. The government thus provides substantial subsidies for the contribution ranging from 40% to 100%, based on the participants' economic and health conditions. Table 1 sets out the percentage of government subsidies to different groups and the definition (and abbreviated form) for all eight groups within the NP; these are the groups included in the study.

The general public (GP) who do not belong to any disadvantaged group are subsidized at a rate of 40% of the premiums. The subsidization rate for citizens with a mild mental or physical disability (D1) and income less than two times the lowest living expense (LOW2.0) is 55%, while the rate is 70% for citizens with a moderate mental or physical disability (D2) and less than moderate to low income (LOW_M) or 1.5 times the lowest living expense (LOW1.5). The rate for citizens with a severe mental or physical disability (D3) and low income (LOW) is 100%. The rate in each case depicts the relative degree of disadvantage: The higher the subsidy percentage rate, the greater the degree of disadvantage.

These subsidy definitions provide precise information on the degree of disadvantage assigned to each category. We use the GP as the comparison reference group and refer to the other groups as the disadvantaged groups. Our purpose is to examine how the implementation of the NP affects the distribution of pension benefits between the GP and disadvantaged groups.

Participants who reach the age of 65 are entitled to a monthly old-age pension with no minimum contribution period required. The NP's monthly pension benefit can be calculated in two ways:

$$A = 0.65\% \cdot \text{Insurance Period(in Year)} \cdot MIA + NPG \quad (1)$$

$$B = 1.3\% \cdot \text{Insurance Period(in Year)} \cdot MIA \quad (2)$$

The monthly pension benefit under the NP is obtained by maximizing Equations (1) and (2), where Equation (1) is based on a general defined benefit (DB) equation, and Equation (2) provides the old-age basic guaranteed pension payment (NPG) for those with short contribution periods. The pension authority calculates the benefit under both equations and pays the participant at the rate that favours their situation.

Equations (1) and (2) both depend on the insurance period (in years). Insured citizens with short participation periods benefit from Equation (1). However, as that participation extends, the likelihood of being entitled to Equation (1) will decrease. Those insured under the NP who have received other social benefits or did not repay their delayed

Table 1. Definitions, subsidy percentages, and abbreviations of NP subsidy identities.

Subsidy Identity	Abbreviation	Subsidy Percentage
General public	GP	40%
Disability Relevant		
Mild mental/physical disability	D1	55%
Moderate mental/physical disability	D2	70%
Severe mental/physical disability	D3	100%
Poverty Relevant		
Income less than 2 times the lowest living expense	LOW2.0	55%
Income less than 1.5 times the lowest living expense	LOW1.5	70%
Moderate Low income	LOW_M	70%
Low income	LOW	100%

premiums within a 10-year grace period are not entitled to the benefits of Equation (1); this is referred to as the 'exclusivity of Equation (1)'.

Because the NP was implemented quite recently, in 2008, current participants have not been able to accumulate contributions over that short period, so most beneficiaries of old-age pensions receive their benefits based on Equation (1).

Citizens can move between the NP and the LI. These citizens can be entitled to their NP and LI pensions after age 65. The monthly pension benefit of the LI, like the NP, has two types: Equations (3) (A^{LI}) and (4) (B^{LI}):

$$A^{LI} = 0.775\% \cdot \text{Insurance Period(in Years)} \cdot MIS + LIG \quad (3)$$

$$B^{LI} = 1.55\% \cdot \text{Insurance Period(in Years)} \cdot MIS, \quad (4)$$

where *MIS* represents the average 60-month MIS (monthly insured salary) of LI, and *LIG* is the basic guaranteed LI pension payment amounting to NT\$3,000.

4. Data analysis

4.1. Data

We use administrative data from the Bureau of Labor Insurance, the state authority that operates the NP programme. The data span from October 2008 through December 2020 and consists of information on the 10,599,518 citizens who currently qualify or have previously qualified for the NP. The citizens in our sample compose about 59% of Taiwan's population aged 25 and older.² Each month, NP-qualified citizens are automatically enrolled in the scheme if they are not otherwise covered by occupational insurance. The data also contain each citizen's subsidy identity, premium payment status (on time, delayed, or unpaid), payment dates, applicable annuity benefit equation (Equations (1) or (2)), and corresponding monthly annuities for citizens over 65. This administrative dataset consists of 565,372,433 observations over 12 years, each representing a citizen who qualified for the NP for a particular month, regardless of whether that citizen paid the premium.

Notably, each NP beneficiary's subsidy identity is recorded monthly and may be subject to change. We ensure the consistent identity of each citizen in our data by excluding those whose identities have changed. The empirical results of our study are based on the subsidy identity, which characterizes not only its degree of disadvantage but also premium-paying behaviours. We also use the subsidy identity to examine the income redistribution among different identity groups. If the identity is changeable, then the effect of income redistribution is difficult to identify. In addition, because we focus on premium contributions and annuity benefits in each citizen's lifetime, we removed from the sample of citizens in the NP who died before the end of the observation period (December 2020). LOW_M is a new subsidy identity established after 2020; the group contains only 1,526 citizens and is excluded from the sample. Around 11% of NP citizens were ultimately excluded from our sample, leaving a total sample size of 9,388,281 citizens (4,594,939 men and 4,793,342 women).

Table 2 presents the number of NP citizens in each of the seven subsidy identity groups for both genders. The GP overwhelmingly outnumbers the other subsidy identities, accounting for 96.51% of the total NP population. The three disability groups and three poverty groups (i.e. the disadvantaged groups) cumulatively account for 2.76% and 0.73% of the entire NP population, respectively. Despite their small sample shares, the total population of the disadvantaged groups is nevertheless sizable at 327,834. Table 2 shows that women outnumber men in the GP and poverty groups, whereas men outnumber women in the disability groups.

4.2. Population and socioeconomic profile by subsidy identities

In addition to presenting the NP subsidy identities in Table 2, we link the NP administrative data with the other three administrative data sets, including year 2020 taxable-income data from the Ministry of Finance's Fiscal Information Agency, LI data from the Ministry of Labor's Bureau of Labor Insurance, and SWA data from the Ministry of Health and Welfare. The merged data provide valuable information for each NP enrollee, including annual personal and household income, LI coverage, and SWA received.

Table 3 shows the mean and standard deviation of annual personal and household income for the GP and disadvantaged groups in 2020. All disadvantaged groups have less personal and household income than those in the GP. Among those in the disability groups, the more severe the disability, the lower the individual and household income; similarly, among the poverty groups, the poorer the group, the lower their income. For

Table 2. Numbers of NP citizens of subsidy identity groups for both genders.

Subsidy Identity	Male	Female	Both genders
GP	4,414,553	4,645,894	9,060,447
Disability			
D1	56,114	34,712	90,826
D2	47,325	34,847	82,172
D3	49,062	37,355	86,417
Poverty			
LOW2.0	4,875	6,968	11,843
LOW1.5	13,477	22,447	35,924
LOW	9,533	11,119	0,652
All Subsidy Identity	4,594,939	4,793,342	9,388,281

Table 3. Mean and standard deviation of personal income and household taxable earnings in the year 2020 of each subsidy identity group (NT\$).

Subsidy Identity	Person Income (Mean)	Person Income (S.D.)	Household Earning (Mean)	Household Earning (S.D.)
GP	340,674	1,503,651	522,392	1,143,187
D1	175,641	477,964	493,317	973,203
D2	114,031	1,169,894	465,652	1,100,021
D3	65,847	386,504	402,480	1,094,545
LOW2.0	194,717	357,577	408,015	604,834
LOW1.5	147,902	261,782	191,242	398,183
LOW	115,785	217,014	67,259	213,261
All	333,148	1,483,151	518,098	1,137,643

both the disabled and poverty groups, the higher the subsidy, the lower the personal and household income.

4.3. Premium contribution by subsidy identities

Because the NP pension depends on the number of months a qualified citizen paid their premium, we follow Chang et al. (2023) and define two terms to measure premium-paying behaviours: premium payment rate (PPR) and qualification intensity (QI). These terms use the number of months of payments (instead of the premium payment amount) to avoid the distorting effect of frequent premium rate adjustments. The PPR is defined as the number of months when a qualified citizen has paid their premiums (Np) divided by the total number of months the citizen qualified for the NP (TQ). This relationship is expressed by Equation (5).

$$PPR = Np/TQ \quad (5)$$

$$QI = TQ/TE \quad (6)$$

$$PCI = Np/TE \quad (7)$$

Columns 1 and 2 in Table 4 present the average QI and PPR for the GP and the six disadvantaged groups; QI is lower for the GP than for most disadvantaged groups. Within the disability groups, the more severe the disability, the higher the value of QI. Similarly, the poverty level and QI are positively correlated. For example, the value of QI is higher for those in the LOW group than in the LOW2.0 and L-LICM 1.5 groups, indicating that those in the more disadvantaged groups have a relatively greater need for the NP.

The GP exhibits a lower PPR than the disadvantaged groups, implying that the GP is less willing to participate in the NP than other groups. As a result of the subsidy, the PPR

Table 4. Averaged QI, PPR, NP, and LI periods of each subsidy identity group.

Subsidy Identity	QI	PPR	PCI = QI*PPR	NP insurance period	LI insurance period
GP	42%	50%	21%	1.83	13.21
D1	51%	57%	29%	2.53	14.28
D2	71%	73%	51%	4.79	10.50
D3	87%	100%	87%	8.59	6.97
LOW2.0	33%	81%	26%	2.92	15.06
LOW1.5	59%	83%	49%	5.79	13.28
LOW	62%	100%	62%	5.17	12.25
All Subsidy Identities	43%	52%	22%	1.94	13.15

Note: The number of observed citizens is 8,120,339. QI is the average of each group weighted by total exposure (TE), and PPR is each group weighted by the total number of qualified months (TQ). The QI, PPR, and PCI formulas refer to Equations (5)–(7). NP and LI insurance periods are the average periods of each group.

for the D3 and LOW groups is 100%. Nevertheless, D2, with a higher subsidy percentage (70%) shows a higher PPR than D1. Within the poverty groups, PPR is slightly higher for LOW1.5 (83%) than LOW2.0 (81%). Therefore, the subsidy is positively associated with the PPR.

The three right-most columns in Table 4 are the PCI, average NP, and LI insurance periods (in years), respectively, for each subsidy identity.³ On average, the NP insurance period is shorter for the GP than for the disadvantaged groups. By contrast, the GP and those groups with a low degree of disadvantage exhibit a longer average LI benefit period, showing that they frequently benefit from LI and thus have a relatively low QI. The PCI is strongly positively correlated with the NP insurance period, with a correlation coefficient of 0.967. In addition, QI is strongly negatively correlated with the LI insurance period, with a correlation coefficient of -0.926 .

5. Annuity benefits across disadvantaged groups and generations

NP, LI pension benefits, and SWA differ depending on socioeconomic status and generation. Our empirical design allows us to investigate and showcase these differences, first by showing the actual NP and LI pension benefits and the SWA for citizens older than 65 in the GP and for members of the various disadvantaged groups. It is important to note that in the month immediately following the implementation of the NP, citizens who had never been employed were automatically covered by the NP; this includes those citizens who reached 65 before 2020. In the sample of citizens older than 65, there is an abnormally high proportion of people who had never been employed: In other words, their LI period and pension benefits are zero.

We refer to this phenomenon as the ‘first-month effect’, which lowers the senior sample’s average LI pension benefit and exaggerates the NP contribution. That is, the NP pension benefit as a proportion of total monthly pension benefits is abnormally large. Although the senior sample offers a practical view of the NP as part of the social safety net for current senior citizens, it does not represent the younger generations, for whom the social insurance system must account. We thus predict the NP and LI pension benefits and SWA for citizens younger than 65 in the GP and disadvantaged groups.

5.1. Actual annuity benefit for citizens aged 65 and above

We compare the extent of actual NP pension benefits and SWA accruing to the GP and each disadvantaged group by selecting NP-qualified citizens older than 65 as our sample. We use each individual’s actual data for NP and LI pension benefits and SWA to compute the averages for the GP and each disadvantaged group. The results are presented in Table 5. Note that there are two kinds of LI benefits: lump-sum and pension benefits. The lump-sum benefits are incomparable to other monthly pension benefits or the SWA; furthermore, those entitled to LI lump-sum benefits are not allowed to claim the NP pension after the transition period, which ended in September 2023. We thus exclude senior citizens who received the LI lump-sum benefit, leaving us with a sample of 441,559 senior citizens.

Table 5 shows the average actual pension benefits of the LI and NP and the SWA for the GP and each disadvantaged group. Other than those in D1, those in the disadvantaged

Table 5. Averaged pension benefits for LI, NP, and SWA by subsidy identity group.

Sub. Id.	LI	SWA	NP	TOTAL	NP. PCT
GP	4,818	303	2,456	7,576	32%
D1	5,531	1,612	1,964	9,107	22%
D2	2,532	2,383	2,636	7,550	35%
D3	1,077	2,716	2,196	5,988	37%
LOW 2	2,984	1,419	2,781	7,184	39%
LOW1.5	2,391	1,948	3,134	7,474	42%
LOW	1,252	5,684	1,757	8,692	20%
All Sub. Id.	4,664	451	2,446	7,561	32%

Note: The number of insured citizens is 441,559. The average benefits of *LI*, *NP*, and *SWA* are divided by the total number of observed citizens in each group. *TOTAL* represents the total monthly benefits amount via summation of *LI*, *NP*, and *SWA* averaged benefits. *PCT* is the percentage of *NP* pension benefit on the total monthly pension amount, *TOTAL*.

groups are entitled to lower *LI* pension benefits than the GP; those in the D3 and LOW groups are entitled to the lowest *LI* pension benefit, reflecting their vulnerability in the labour market. However, after taking into account the *NP* pension benefit and the *SWA*, the total monthly pension benefits to disadvantaged groups (except D3) are close to or higher than the GP. Specifically, most of the total benefits of the LOW and D3 groups come from the *SWA*. Unlike these two groups, the *NP* pension benefits of other disadvantaged groups amount to more than the *SWA*.

The last column of Table 5 (*NP. PCT*; percentage of the *NP* pension benefit to the total monthly benefits) shows that the *NP* pension benefit for the GP is 32% of total monthly pension benefits. This proportion is relatively low for the LOW and D1 groups, at 20% and 22%, respectively. The *SWA* amount is based on the degree of disability or poverty experienced by the individual. In contrast, *NP* pension benefits are based on the number of months that a citizen has paid the *NP* premium and their eligibility for Equation (1). Although the last column of Table 5 depicts the *NP* contribution for citizens older than 65, recall that the first-month effect exaggerates the *NP* contribution to the social safety system. Next, we predict the *NP* contribution to the young generation.

5.2. Predicted annuity benefit for citizens aged below 65

Table 5 reflects the actual *NP* and *LI* pension benefits and the *SWA* for citizens over 65. However, it is also important to probe the future *NP* and *LI* pension benefits, as well as the *SWA*, for citizens younger than 65 in the GP and in each disadvantaged group. We predict these using individual-level simulations based on each citizen's *NP*, *LI*, and *SWA* records, as well as their *NP*-premium-paying behaviours.

The simulation procedure is summarized as follows. First, we apply each citizen's past *QI* and $1-QI$ to their remaining years of *NP* and *LI* qualification (until age 65), respectively. The *NP* insurance period is then obtained by multiplying each citizen's past *PPR* by their predicted future *NP* qualification period (future *TQ*). The *PPR* for *LI* is 100% because this premium is mandatory. We obtain the total insurance periods of the *NP* and the *LI* by adding the past to the predicted insurance periods. We further compute the corresponding future pension benefits based on the DB equations of *NP* (Equations (1) and (2)) and *LI* (Equations (3) and (4)), respectively.

We follow Chang et al. (2023), who use the 2019 NP Actuarial Report, by applying the MIA of the NP, which is NT\$18,282. We infer the future LI monthly insurance salary (hereinafter *LI MIS*) individually based on the average salary adjustment rate over the working period.⁴ The last period of actual SWA for each citizen is used to predict the future SWA. Note that citizens not entitled to the LI and SWA are eligible for NP Pension Benefit A (Equation (1)) if they pay all NP premiums; that is, PPR = 100%. This reflects the ‘exclusivity of Equation (1)’ mentioned above.

Table 6 shows the predicted LI pension benefit, SWA, and NP pension benefit of the GP and different disadvantaged groups for three age cohorts: 35–40, 45–50, and 55–60. The GP are not entitled to the highest total. The total monthly pension benefits are greater for those in the LOW group than for the GP in the 45–50 and 55–60 age cohorts and close to the total monthly pension benefits of the GP for the 35–40 age cohort. Without the NP benefits, the LOW group’s pension benefits from the LI and SWA are less than those of the GP. However, after incorporating the NP benefits, the total monthly benefits of the LOW group are higher than or close to those of the GP.

The last column shows that the higher an individual’s degree of disability or poverty, the higher the share of NP pension benefits in their total monthly benefits. Group D3 (those with the most severe disabilities) has the highest proportion of NP pension benefits among all

Table 6. Predicted LI pension benefit, SWA, NP pension benefit, total monthly benefits, and percentage of NP pension benefit by subsidy identity and age cohort.

Subsidy ID	LI	SWA	NP	TOTAL	NP. PCT
Age Cohort: 35–40					
GP	18,646	6	1,372	20,024	7%
D1	15,645	1,866	2,142	19,653	11%
D2	10,461	2,848	4,068	17,377	23%
D3	5,322	2,611	7,741	15,674	49%
LOW2.0	19,683	17	1,715	21,416	8%
LOW1.5	14,282	100	3,268	17,650	19%
LOW	11,445	2,592	5,443	19,480	28%
ALL	18,505	48	1,430	19,983	7%
Age Cohort: 45–50					
GP	16,434	13	1,309	17,755	7%
D1	15,506	1,775	1,675	18,956	9%
D2	10,034	2,754	3,548	16,336	22%
D3	6,003	2,861	6,016	14,880	40%
LOW2.0	17,114	23	1,734	18,871	9%
LOW1.5	11,051	86	3,520	14,657	24%
LOW	12,908	2,756	3,671	19,335	19%
ALL	16,234	79	1,396	17,709	8%
Age Cohort: 55–60					
GP	14,027	20	1,193	15,240	8%
D1	14,517	1,914	1,188	17,618	7%
D2	10,961	2,748	2,178	15,888	14%
D3	7,037	3,227	3,972	14,237	28%
LOW2.0	12,964	30	1,701	14,695	12%
LOW1.5	11,142	152	2,499	13,793	18%
LOW	11,553	3,767	2,547	17,867	14%
ALL	13,864	142	1,260	15,265	8%

Note: The total number of observed citizens is 8,120,339, denoted by *ALL*. *LI*, *SWA*, and *NP* are the average benefits for each group and age cohort. *TOTAL* is the summation of *LI*, *SWA*, and *NP* for the total monthly benefit. The low *SWA* of GP and *ALL* corresponds to their low claim rate.

subsidy identities. Within the disability groups, the more severe the disability, the lower the total monthly and LI pension benefits, suggesting their low employment capacity. Although there is no apparent relationship between the poverty level and total monthly benefits, the share of the NP pension benefit positively correlates with the degree of poverty.

In Table 6, we observe that the overall NP pension percentage of total monthly benefits is 7% to 8% for citizens younger than 60. These proportions are substantially smaller than the 32% in Table 5, demonstrating the NP's exaggerated contribution for senior citizens above 65, reflecting the 'first-month effect'. Note that the substantial drop in the overall NP pension benefit proportion for the GP. These proportions increase, however, even for the younger among the severely disabled, D3 and LOW groups, suggesting that the NP is more important to the younger members of the severely disabled group than to the GP.

6. Regressions of pension benefits

In this section we investigate the relationships between different types of pension benefits, demographic characteristics, and degrees of disadvantage, using the predictions of each individual's pension benefits and the SWA (from Section 5) to conduct regression analyses of the total monthly pension benefits, LI pension benefits, and the SWA. We also investigate whether the poor receive relatively more pension benefits and SWA than the rich. We explore these relationships using absolute and relative measures and two models, with and without considering premium-paying behaviours, as shown below:

$$y = X\alpha + D\beta + L\gamma + \varepsilon \quad (8)$$

$$y = X\alpha + D\beta + L\gamma + P\rho + \varepsilon \quad (9)$$

In Equations (8) and (9), y is either an absolute or relative measure of pension benefits; X is a vector containing sex, age, and income variables, and α is a vector of the coefficients of the variables in X . D is a vector of three disability variables: $D1$, $D2$, and $D3$. L is a vector of three low-income variables: $LOW2.0$, $LOW1.5$, and LOW . β and γ are regression coefficient vectors of D and L , respectively. P includes three premium-paying behaviours (QI , PPR , and PCI) and ρ is a vector of coefficients of the variables in P ; ε is the regression error term.

6.1. Absolute measure

We first present the results for the absolute measure. Because the NP pension is equal to the total monthly benefit (Model 1 in Table 7) minus the sum of the LI pension benefit and the SWA (Models 2 and 3 in Table 7), the three models of dependent variables in Table 7 capture the whole figure of the construction of pension system for most Taiwanese. Because the monthly NP benefit is determined by the premium-paying behaviour (QI , PPR , and PCI), regression of the NP monthly benefit on the premium-paying behaviour is not meaningful. Moreover, because total monthly benefits = NP monthly benefit + LI pension benefit + SWA, the NP monthly benefit is perfectly associated with the other three. Hence, we do not conduct the regression of the monthly NP benefit. In Table 7, Model 1–1 employs total monthly pension benefits as the dependent variable with independent

Table 7. Regression models for total monthly benefits, LI pension benefit, and SWA.

	y=total monthly benefits		y=LI pension benefit		y=SWA	
	Model 1-1	Model 1-2	Model 2-1	Model 2-2	Model 3-1	Model 3-2
Intercept	21874.43 1738.86 0***	27003.24 3159.27 0***	19183.78 1407.48 0***	24725.40 2799.21 0***	176.84 198.52 0***	191.53 181.58 0***
Female	-190.32 -31.92 0***	-462.77 -134.80 0***	-517.00 -80.04 0***	-452.86 -127.65 0***	0.61 1.44 0.15	2.45 5.78 0***
AGE	-106.75 -380.92 0***	-88.30 -544.53 0***	-75.88 -249.90 0***	-33.53 -200.06 0***	-3.18 -160.08 0***	-3.07 -153.48 0***
D1	644.14 19.51 0***	1546.19 81.65 0***	-1764.02 -49.32 0***	-367.60 -18.78 0***	1848.07 790.64 0***	1848.35 790.93 0***
D2	-1174.79 -34.49 0***	1865.75 95.21 0***	-5922.74 -160.49 0***	-1066.62 -52.67 0***	2790.30 1156.89 0***	2791.14 1154.19 0***
D3	-2244.62 -68.79 0***	1159.69 60.95 0***	-9597.55 -271.46 0***	-1995.84 -101.50 0***	2892.48 1251.80 0***	2900.17 1235.05 0***
LOW2.0	954.30 9.42 0***	-1388.10 -23.89 0***	525.68 4.79 0***	-1441.00 -23.99 0***	22.23 3.10 0**	30.45 4.25 0***
LOW1.5	-2103.19 -43.13 0***	-1147.94 -41.00 0***	-3969.03 -75.13 0***	-1347.76 -46.58 0***	97.87 28.34 0***	102.97 29.80 0***
LOW	1780.12 28.48 0***	2076.72 57.82 0***	-4384.14 -64.74 0***	-1498.78 -40.38 0***	3459.61 781.64 0***	3470.43 783.03 0***
Income	9.60 442.77	2.96 235.89 0***	10.30 438.50	3.15 242.99 0***	-0.05 -32.15	-0.04 -22.73 0***
QI		-19460.50 -2442.91 0***		-19615.80 -2382.75 0***		7.43 7.56 0***
PPR		2133.34 338.51 0***		1881.08 288.83 0***		-40.30 -51.82 0***
PCI		6102.27 594.39 0***		-337.70 -31.83 0***		6.81 5.38 0***
R ²	4%	69%	5%	71%	33%	33%

Note: The sample size is 8,120,339. For each independent variable, values in the first, second, and third rows are coefficients, t-values, and p-values; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

variables including *Female* (men being the reference group), *AGE*, disadvantaged group (the GP as the reference group), and personal income (*INCM*).

To examine the effect of the NP premium-paying behaviours on pension benefits, we add *QI*, *PPR*, and *PCI* in Model 1–2. *PCI* can also represent the effect of long-term qualification and the intensity of NP premium payments. We also use the LI pension benefit as the dependent variable in Models 2–1 and 2–2 and the *SWA* as the dependent variable in Models 3–1 and 3–2, with the same independent variables as in Models 1 and 2, respectively.

In Models 1–1 and 1–2, the negative coefficients of *Female* show that the total monthly benefits are lower for women than for men. This is because women receive fewer LI pension benefits, as shown in Models 2–1 and 2–2 (*Female* coefficients are negative in both models). However, as demonstrated in Models

3–1 and 3–2, women receive more SWA. The results for *Female* indicate that, compared to men, women are less likely to work (or, if they do work, to have lower earnings), and also that they are more likely to be disadvantaged. The coefficients of *AGE* are negative in all models in Table 7. Senior citizens usually have a lower LI MIS and a lower LI pension benefit (Models 2–1 and 2–2). Moreover, because the scheme only began in 2008, senior citizens do not have a long period of NP participation.

Without considering the *QI*, *PPR*, and *PCI*, the total monthly benefits to disadvantaged groups D2, D3, and LOW1.5 are less than in the GP (Model 1–1). This is because their LI pension benefit is much less than the SWA (Models 2 and 3). Taking into account the *QI*, *PPR*, and *PCI*, the coefficients of *LOW1.5* and *LOW2.0* are significantly negative in the total monthly pension benefits regression. This suggests that, given the NP premium-paying behaviours, these two disadvantaged groups receive lower total monthly pension benefits than the GP. These results are unsurprising because these groups are not officially defined in the Public Assistance Act, and their members are not eligible to receive the SWA (small coefficients of *LOW2.0* and *LOW1.5* in Model 3).

The higher the personal income (*INCM*), the higher the total monthly benefits (Models 1–1 and 1–2) and LI pension benefit received (Models 2–1 and 2–2), but the lower the SWA (Models 3–1 and 3–2). The results of Model 1–2 indicate that the total monthly benefits are negatively associated with *QI* (Model 1–2). This suggests that citizens with greater rates of inclusion in the NP (high *QI*) exhibit low employment intensity. Consequently, their LI pension benefit is low (Model 2–2), and their total monthly benefits are also low (Model 1–2). The positive coefficient of *PCI* implies that citizens contributing more in premiums to the NP are entitled to greater NP pension benefits (Model 1) and, hence, to greater total monthly benefits but a lower LI pension benefit (Model 2).

The coefficient of *PCI* is also positive in the SWA model. The *PCI* coefficients in Models 1 and 2 take the opposite signs, indicating that the NP and the LI are substitute goods, whereas the same sign for *PCI* in Models 1 and 3 suggests that the NP and the SWA are complementary goods. Citizens with a lower LI pension benefit will be compensated within the social safety net with a higher NP pension benefit. A higher NP pension benefit will supplement the benefits of citizens with a higher SWA.

In sum, although women, the poor, and the elderly are entitled to more social welfare allowance, they are entitled to far lower LI pension benefits. As a result, they receive less total monthly benefits than their counterparts. After controlling for the intensity of the NP's inclusion and the NP's premium-paying behaviour, all disadvantaged groups officially defined in the Public Assistance Act receive more total monthly benefits than the GP. Furthermore, the more frequently they pay the NP premium, the greater the total monthly benefits they will be entitled to. This demonstrates the crucial role of the NP in the redistribution of income.

6.2. Relative measure

In addition to the absolute measures in Table 7, we investigate the relative importance of the NP benefit, the LI pension benefit, and the SWA for the GP as well as each disadvantaged group using relative measures. That is, the dependent variables are the proportions of the NP benefit (*NP%*), LI pension benefit (*LI%*), and the SWA (*SWA%*) to total monthly

Table 8. Regression models for percentages of NI, LI, and SWA.

	y=NP %		y=LI %		y=SWA %	
	Model 4-1	Model 4-2	Model 5-1	Model 5-2	Model 6-1	Model 6-2
Intercept	0.1340 411.8774 0***	0.0934 436.8006 0***	0.8550 2600.5899 0***	0.8990 4069.7957 0***	0.0110 169.6872 0***	0.0076 99.5170 0***
Female	0.0332 215.5799 0***	0.0057 66.0156 0***	-0.0333 -213.7983 0***	-0.0061 -69.3516 0***	0.0001 2.3783 0.02*	0.0005 15.8025 0***
AGE	-0.0006 -82.2664 0***	-0.0026 -648.8924 0***	0.0008 108.7054 0***	0.0028 675.0607 0***	-0.0002 -138.5316 0***	-0.0002 -136.1980 0***
D1	0.0290 34.1516 0***	-0.0141 -29.9840 0***	-0.1660 -193.4267 0***	-0.1226 -251.9629 0***	0.1370 810.1057 0***	0.1367 811.7331 0***
D2	0.1480 168.6403 0***	-0.0132 -27.0038 0***	-0.3599 -405.8580 0***	-0.1982 -392.9880 0***	0.2119 1212.7762 0***	0.2114 1210.8175 0***
D3	0.3785 452.0708 0***	0.0201 42.6387 0***	-0.5682 -671.4795 0***	-0.2107 -431.9089 0***	0.1897 1137.4122 0***	0.1906 1128.5676 0***
LOW2.0	0.0232 8.9293 0***	0.0024 1.6639 0.1	-0.0243 -9.2516 0***	-0.0050 -3.3223 0***	0.0011 2.1084 0.03*	0.0026 4.9464 0***
LOW1.5	0.1530 122.2357 0***	0.0135 19.4231 0***	-0.1598 -126.3352 0***	-0.0208 -28.9664 0***	0.0068 27.2793 0***	0.0073 29.3839 0***
LOW	0.1511 94.2260 0***	-0.0619 -69.5028 0***	-0.3431 -211.7124 0***	-0.1316 -142.9075 0***	0.1920 601.2165 0***	0.1935 607.1538 0***
Income	-8.67E-09 -154.4327 0***	-1.73E-09 -54.9926 0***	9.14E-09 161.1215 0***	1.87E-09 57.5493 0***	-4.71E-10 -42.1249 0***	-1.41E-10 -12.5218 0***
QI		0.0645 318.3516 0***		-0.0767 -366.7075 0***		0.0123 169.4286 0***
PPR		-0.0036 -23.0653 0***		0.0038 23.1058 0***		-0.0001 -2.2708 0.02*
PCI		0.5335 2061.2283 0***		-0.5239 -1958.8020 0***		-0.0096 -103.4112 0***
R ²	4%	70%	9%	71%	32%	32%

Note: Citizens with missing or zero total monthly benefits will be ruled out. The sample size is 8,006,919. For each independent variable, values in the first, second, and third rows are coefficients, t-values, and p-values; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

pension benefits, respectively. The results are shown in Table 8 in Models 4–1 and 4–2, Models 5–1 and 5–2, and Models 6–1 and 6–2, respectively.

The coefficients of *Female* in the *NP%* models (Models 4–1 and 4–2) and *SWA%* models (Models 6–1 and 6–2) are significantly positive, whereas in the *LI%* models (Models 5–1 and 5–2), the *Female* coefficient is significantly negative. The former implies that the NP benefit and the SWA are more important to women than to men; the latter suggests that women are entitled to relatively fewer LI pension benefits than men. Thus, implementing the NP narrows the pension benefit gap between the sexes.

The *AGE* coefficients in the *NP%* models (Models 4–1 and 4–2) and *SWA%* models (Models 6–1 and 6–2) are negative, indicating the greater importance of the NP and SWA for the young. The positive coefficient of *AGE* in the *LI%* models (Model 5–1 and 5–2)

shows that the LI pension benefit is more important for the elderly. The coefficients of *INCM* are negative in the *NP%* and *SWA%* models and positive in the *LI%* models. These results suggest that the higher an individual's income, the lower their NP pension benefits and SWA proportion, and the higher the LI pension benefit as a share of the total monthly benefits. Hence, the NP and the SWA are more important to the poor than the rich.

Without considering the NP premium-paying behaviours (i.e. *QI*, *PPR*, and *PCI*), all coefficients are positive for the disadvantaged groups, indicating that the NP pension benefit (Model 4–1) accounts for a higher proportion of total monthly pension benefits for those in disadvantaged groups than for the GP. In contrast, the GP enjoys a higher LI pension benefit than disadvantaged groups (Model 5–1). After incorporating NP premium-paying behaviours, the coefficients of the disadvantaged groups substantially decrease in the *NP%* models (Model 4–1 vs Model 4–2). Citizens not covered by the NP are likely to be covered by the LI. Therefore, *QI*, *PPR*, and *PCI* reflect citizens' work status, and the signs of their coefficients are opposite in the *NP%* and *LI%* models. The substantial decrease in the coefficients for the disadvantaged groups implies that the NP premium-paying behaviours have strong explanatory power in the *NP%* model. Adding the NP premium-paying behaviours markedly enhances the models' explanatory power (R^2). In other words, the participation and premium payments to the NP are crucial to explaining the importance of the NP and LI pension benefits as a share of total monthly pension benefits.

The coefficient of *PCI* is positive in the *NP%* model (Model 4–2) and negative in the *LI%* model (Model 5–2), reflecting that NP and LI pension benefits are relative substitutes. *QI* measures the intensity of NP qualification: The higher the value of *QI*, the more often a citizen is included in the NP; hence, *QI* is positively associated with the *NP%* and negatively associated with the *LI%*. The more often a citizen is unemployed, the more likely it is that they will be entitled to the SWA. This explains the positive coefficient of *QI* in the *SWA%* model. Although the sign of the coefficient of *PPR* is the opposite of the *PCI* coefficient in the *NP%* and *LI%* models, the magnitude of the latter is far greater.

Similarly, the positive coefficient of *PCI* in the *NP%* model and the negative coefficient of *PCI* in the *SWA%* model imply that the relationship between NP pension benefits and SWA for citizens contributing a high premium to the NP is relatively substitutive. It may appear contradictory that the SWA is a complementary good in absolute terms whereas it is a substitute good in relative terms. This is because, as seen in [Table 7](#), although the coefficient of *PCI* is very small in the *SWA%* model, it is large in the total monthly benefit, suggesting that a 1% increase in *PCI* increases the NP benefit much more than it increases the SWA. Because the *SWA%* denominator is the sum of the pension benefits (including the NP benefit) and the SWA, and its numerator is the SWA, the *PCI* increases the denominator much more than it increases the numerator. Therefore, the coefficient of the *PCI* in the *SWA%* model is negative, demonstrating that the NP and the SWA are relative substitutes.

In sum, this section demonstrates that for disadvantaged groups, the NP is more important than other pension benefits. It further suggests that the implementation of the NP benefits disadvantaged groups and narrows the income gap between

disadvantaged groups and the GP when they age. The substitute relationship between the NP% and the LI% reinforces that the NP compensates people who did not have a job when they were young. As expected, the more premiums people contribute to the NP, the greater the importance of the NP component in their total monthly pension.

7. Conclusions and policy implications

7.1. Conclusions

The purpose of the NP is to help citizens excluded from occupational social insurance to enjoy a better quality of life in old age. Many of these citizens are disadvantaged, and, as is well known, their life expectancy therefore tends to be shorter. A social insurance programme with a unified premium rate is unfair to such individuals. This study examines whether the implementation of the NP reduces the income gap in favour of the disadvantaged under the NP scheme in particular and the social safety net as a whole. Moreover, we explore whether the NP helps disadvantaged individuals in the younger generation more than those in the older generation.

Our study investigates the above issues using NP, LI, SWA, and taxed-income administrative data. In the short term, the SWA is more critical than the NP pension benefit to the elderly (over 65) who are most severely disadvantaged (D3 and LOW). In the long term, the NP is more important than the SWA in reducing the income gap among the older generation in favour of the disadvantaged; this income gap reduction is more substantial for the young than for the old. The NP pension scheme narrows the benefit gap between the disadvantaged and non-disadvantaged groups; this narrowing effect is stronger for the younger than for the older generation.

Finally, in the regression analyses, we find that the NP pension benefit is relatively more important for women and the younger generation than for men and older citizens, and also that it is more important for the poor than for the non-poor. In addition, citizens who make more NP premium contributions receive fewer LI pension benefits but more SWA; these results suggest that the NP and the LI are absolute substitutes. In contrast, the NP and the SWA are absolute complementary goods; the NP is an absolute substitute for the LI and an absolute complement to the SWA. However, the NP and the LI (as well as the NP and the SWA) are substitutes in relative terms. This means that the NP is more important to citizens with less LI coverage and SWA assistance.

7.2. Policy implications

Unlike the other disadvantaged groups, the LOW1.5 and LOW2.0 identities do not afford individuals an entitlement to the SWA. LI pension benefits are insufficient for those in both of these groups, and the NP offers their only additional pension benefit, explaining why they have the highest PPR (except for the 100% subsidization rate). Given the NP-premium-paying behaviours, these two groups have lower total monthly pension benefits than other disadvantaged groups, and the NP pension benefits they receive are more important to them than they are for other disadvantaged groups (except for those with

a severe mental or physical disability). Thus, Taiwanese authorities may consider extending the SWA to citizens in these two groups.

Our results highlight the importance of the NP in providing old-age income protection for Taiwanese citizens not participating in the labour market. In many countries, particularly those with Bismarckian pension systems, the public pension system covers only employed persons and excludes those without paid employment. Under such a system, older people face a higher risk of poverty. In response to this concern, many countries provide a tax-funded, means-tested allowance as supplementary income for older people. In contrast, Taiwan implements a unique contributory pension scheme to cover citizens not participating via occupational groups, which has played a substantial role in providing income protection to that group. It is essential that the Taiwanese government maintain the NP on a long-term, sustainable basis despite its operational challenges. In addition, Taiwan's unique NP model warrants further study of its outcomes and impacts.

This study demonstrates that Taiwanese citizens who enjoy better economic conditions are more likely to pay NP premium contributions and can, therefore, accumulate a higher pension entitlement. By contrast, economically disadvantaged citizens may lose their chance to benefit from such a scheme due to insufficient contribution records. In response, the government may increase subsidies to such individuals to improve their capacity to pay contributions; greater government support is particularly important to them because they do not have employers to pay a share of their contribution. Increased financial support can improve the incentives for participants to pay premium contributions regularly, which is a precondition to earning sufficient old-age pension benefits. Therefore, government subsidies can strengthen the NP and maximize its policy goal of protecting older people from poverty.

Indeed, with more premium subsidies for the poor, the design of the NP must have a redistributive effect that favours them. The redistributive effect can prevail only if NP participants pay the premiums. However, Taiwan's NP faces a critical financial sustainability problem because the young generation is more reluctant to pay the premium than the old generation. However, because our empirical results show that the income-gap reduction is more substantial for the young generation than for the old generation, young NP participants have a greater material incentive to continuously pay the premium. To solve the financial sustainability problem and to mitigate future income inequality in advance, Taiwanese authorities must raise awareness of this reality and encourage the young generation to support the NP.

The findings of this study also echo the value of the multi-pillar pension system advocated by the World Bank (Holzmann & Hinz, 2005; World Bank, 1994). Taiwan's case reveals that the tax-funded SWA is currently an important mechanism for old-age income protection despite the implementation of major social insurance pension schemes. Contributory pension schemes have their limits: Those who are not eligible to participate or are unable to pay contributions are left behind and, therefore, must depend on other mechanisms for old-age income security. The tax-funded SWA can fill these gaps, supporting the argument that a well-run multi-pillar pension system may be the best solution to eliminate old-age poverty.

At least two factors may change income inequality among the elderly in the future. First, as we pointed out in the paper, the first-month effect exaggerates the NP contribution to the social safety system. When the first-month effect disappears, the NP contribution to the social

safety system will decline. Second, we assumed that the same proportion of young generations would continue to pay the NP premium in the future. However, as the proportion of young people paying the NP premium continues to decline, income inequality may worsen.

Empirical studies have shown that the Bismarckian pension system may generate greater income inequality among the elderly population (Ebbinghaus, 2021). Although we demonstrate that Taiwan's NP has a substantial impact on reducing old-age income poverty, it is unreasonable to expect that the NP will totally reduce income inequality among the elderly population. In this regard, a universal, multi-pillar pension system integrating diverse pension schemes may be apt for this purpose. However, as seen in many countries with a mature Bismarckian pension system, the combination of institutional legacies and financial constraints pose extreme obstacles to structural reform (Marier, 2002, Natali & Rhodes, 2004b, Natali & Rhodes, 2004a). Therefore, it is likely that Taiwan's pension system will continue to follow the Bismarckian model for the foreseeable future.

Finally, is Taiwan's NP a good model for other countries with Bismarckian pension models? As indicated earlier, such countries usually disburse tax-funded benefits to those who do not receive sufficient old-age pension. By contrast, Taiwan's NP is a unique pension scheme that covers only those who are not employed based on a contributory principle, and this study has demonstrated its effectiveness in reducing old-age poverty. However, more studies are needed to explore its impact on income adequacy, financial sustainability, and administrative efficiency before discussing the value of Taiwan's NP for cross-national policy learning.

Notes

1. Including Austria, Belgium, Bulgaria, Czech Republic, Cyprus, Estonia, Hungary, Greece, France, Italy, Latvia, Lithuania, Luxembourg, Malta, Poland, Portugal, Romania, Slovakia, Slovenia, and Spain. See Table 2 in Assal et al. (2023).
2. Taiwan's population aged 25 and older was 17,948,531 in December 2020. In 2020, around 3.2 million citizens were currently covered by the NP. In addition, 7.4 million citizens were not currently covered by the NP in 2020 but were covered before 2020. Therefore, 10.6 million citizens will be entitled to the NP pension. This figure is 59% of the Taiwanese population aged 25 years and older.
3. The total number of LI premiums paid during the period of joining LI insurance coverage are also named by insurance period (in years) according to Article 19 of the Labor Insurance Act (LIA).
4. Monthly insurance salary is named in No. 13 of the LIA. LI MIS is subject to a maximum of NT \$45,800 and a minimum NT\$25,600.

Abbreviations

Abbreviation	Full name
NP	National Pension
LI	Labor Insurance
SWA	Social welfare allowance
MIA	Monthly insured amount
MIS	Monthly insured salary
NPG	Old-age basic guaranteed pension payment

LIG	Basic guaranteed pension payment of the LI
PPR	Premium payment ratio
QI	Qualification intensity
PCI	Premium contribution intensity
TQ	Total months a citizen is qualified for the NP
TE	Total exposure (the total number of months a citizen was in the data)
N _p	Total months a citizen paid the premium

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